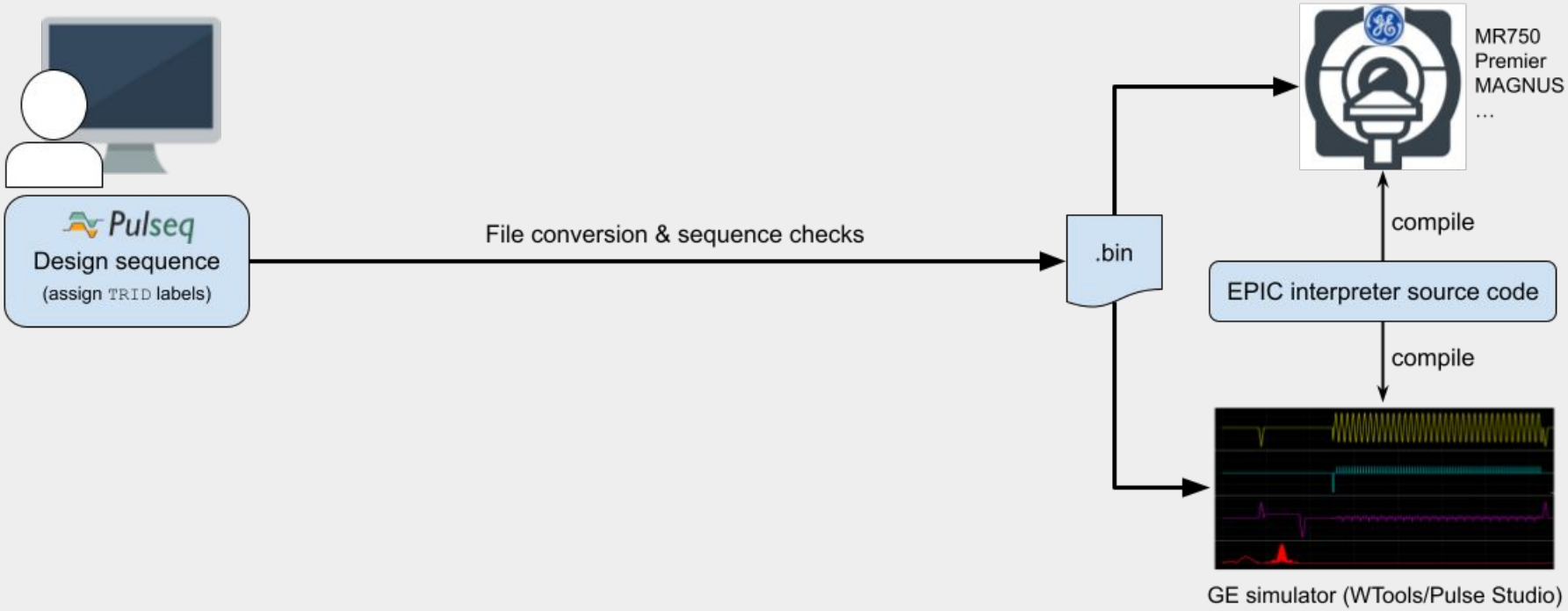
Pulseq interpreter and workflow on GE

Jon-Fredrik Nielsen

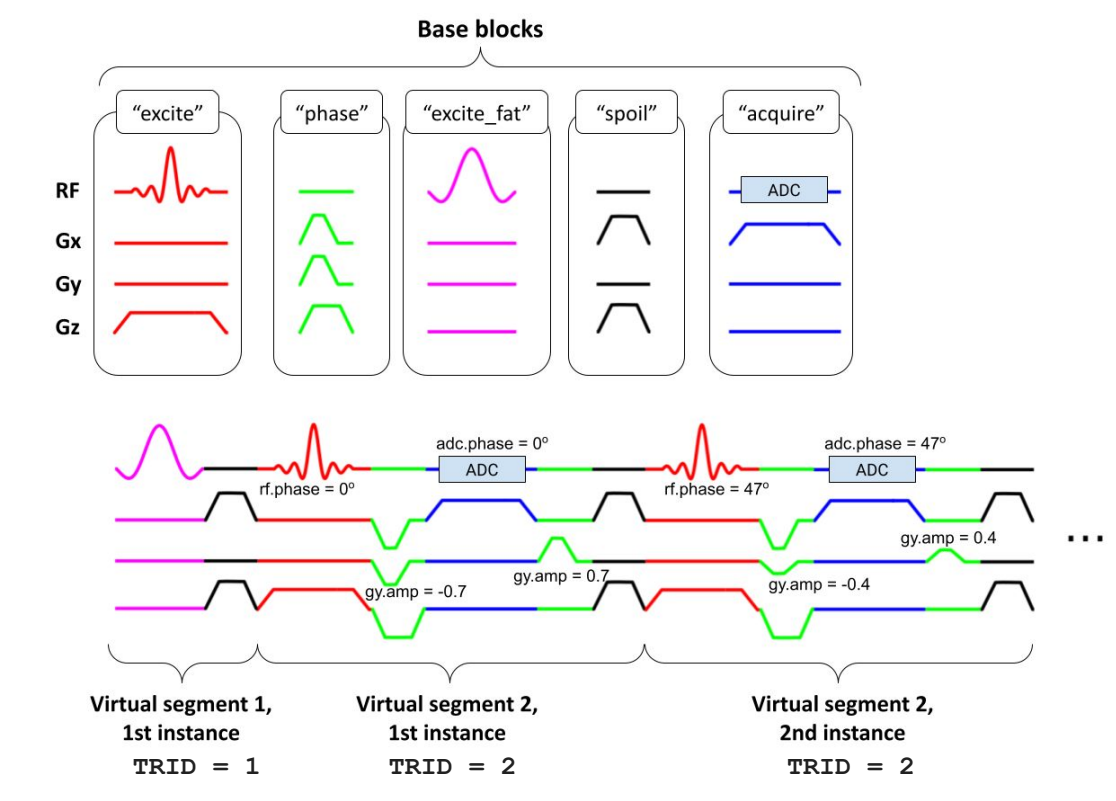
Tues Feb 24, 2026

ISMRM Virtual Meeting: Vendor-Agnostic Pulse Sequence Programming with Pulseq II

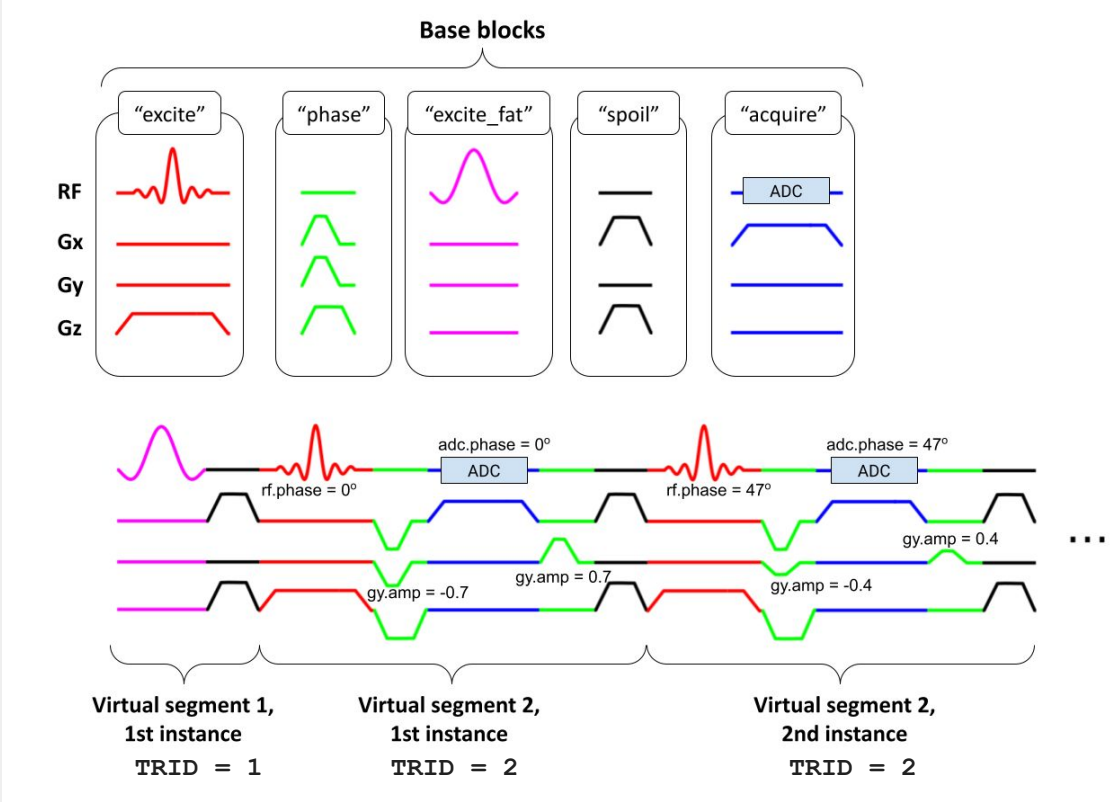
Workflow


Pulseq
Design sequence
(assign TRID labels)
File conversion & sequence checks
.bin
MR750
Premier
MAGNUS
...
compile
EPIC interpreter source code
compile
GE simulator (WTools/Pulse Studio)

GE: Segmented (“TR”) sequence representation



GE: Segmented (“TR”) sequence representation



NEW

PulSeg

https://github.com/HarmonizedMRI/pulseg/

```
struct PulsegSequence
    baseBlocks: [1x8 struct]
    virtualSegments: [1x4 struct]
    scanLoop: [2462x23 double]
end

psq = pulseg.fromSeq('gre.seq');
```

(Formerly “seq2ceq.m” in PulCeq toolbox)

How to assign TRID labels (when creating the .seq file)

Until now:

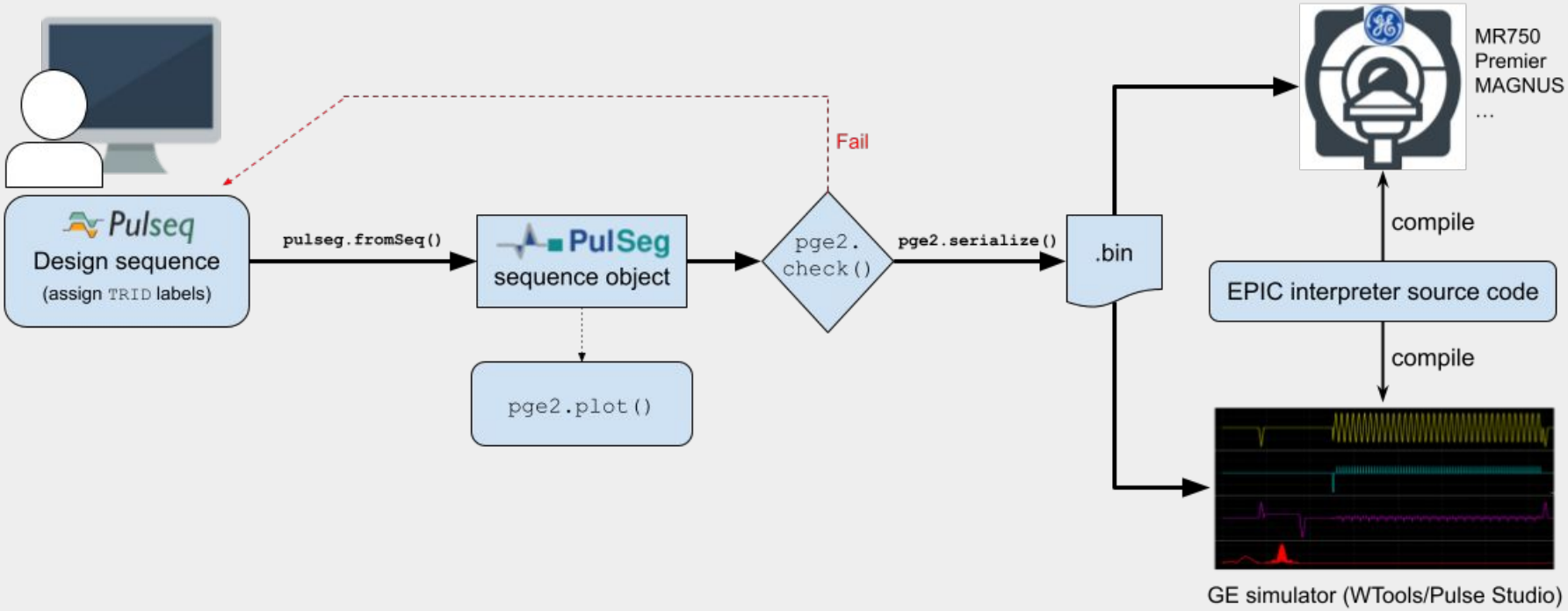
```
>> seq.addBlock(mr.makeLabel('SET', 'TRID', <int_id>));
```

New! (Thanks to Maximilian Gram)

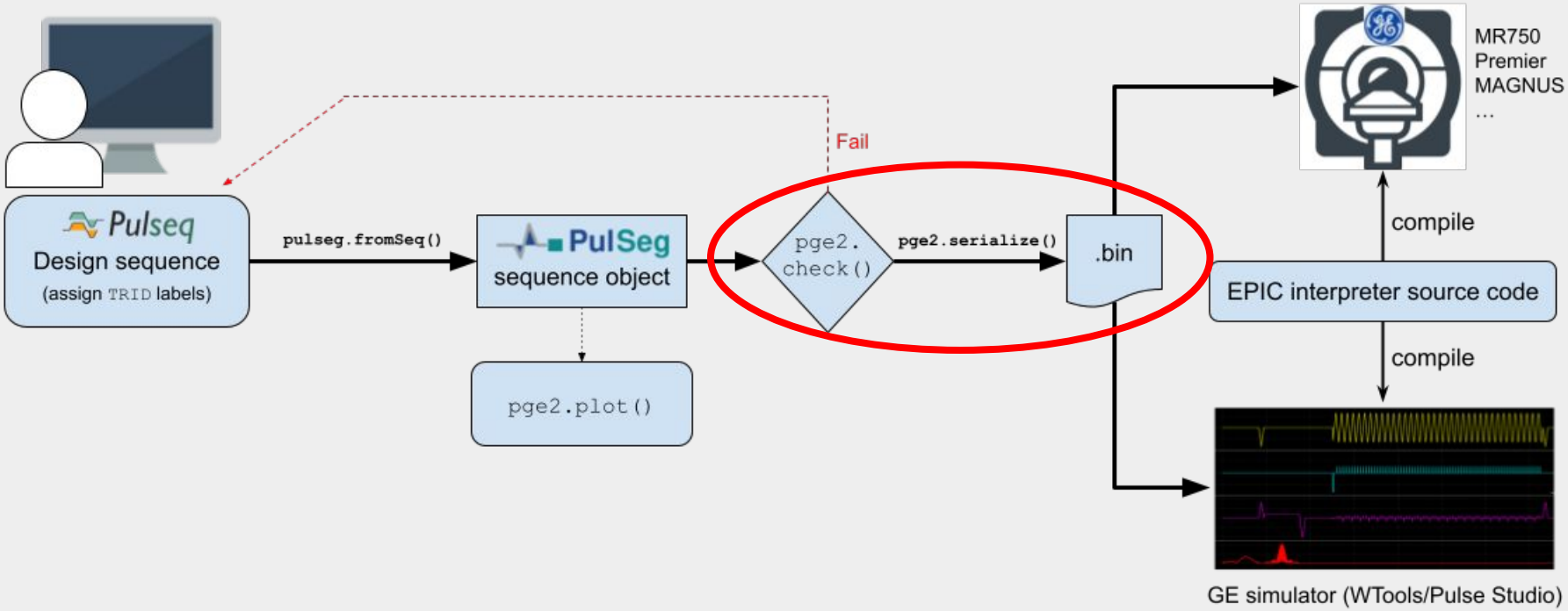
```
>> seq.addTRID("your_label");
```

- No need to keep track of explicit int IDs -- this is handled internally
- Greatly simplifies complex/modular sequence assembly

Workflow


Pulseq
Design sequence
(assign TRID labels)
pulseg.fromSeq()
PulSeg
sequence object
pge2.plot()
Fail
pge2.check()
pge2.serialize()
.bin
MR750
Premier
MAGNUS
...
compile
EPIC interpreter source code
compile
GE simulator (WTools/Pulse Studio)

Workflow


Pulseq
Design sequence
(assign TRID labels)
pulseg.fromSeq()
PulSeg
sequence object
pge2.plot()
Fail
pge2.
check()
pge2.serialize()
.bin
MR750
Premier
MAGNUS
...
compile
EPIC interpreter source code
compile
GE simulator (WTools/Pulse Studio)

Serialize PulSeg object for execution on GE scanners

```
%% Load sequence
psq = pulseg.fromSeq('gre.seq');

%% Define scanner hardware (MR750 example)
psd_rf_wait  = 100e-6;   % RF-gradient delay (s)
psd_grd_wait = 80e-6;    % ADC-gradient delay (s)
b1_max   = 0.25;         % Gauss
g_max    = 5;            % Gauss/cm
slew_max = 20;           % Gauss/cm/ms
coil     = 'xrm';        % Coil name for PNS checks

sysGE = pge2.opts(psd_rf_wait, psd_grd_wait, b1_max, g_max, slew_max, coil);

%% Run PNS and hardware compatibility checks
PNSwt = [0.8 1 0.7];   % directional PNS weights, see pge2.pns()
params = pge2.check(psq, sysGE, 'PNSwt', PNSwt);

%% Export to binary file
pge2.serialize(psq, 'output.bin', 'sysGE', sysGE, 'params', params, 'pislquant', 64);
```

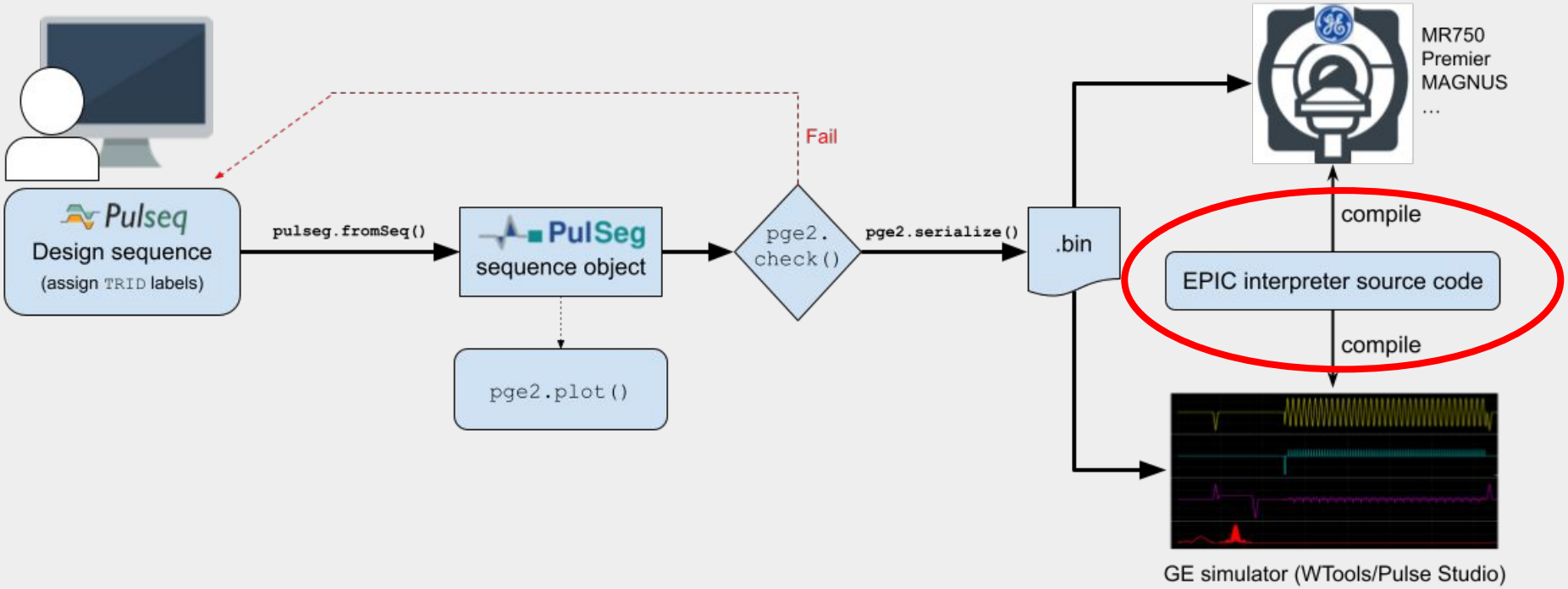
NEW

+pge2

https://github.com/HarmonizedMRI/pge2/

(Formerly part of PulCeq toolbox)

Workflow


Pulseq
Design sequence
(assign TRID labels)
pulseg.fromSeq()
PulSeg
sequence object
pge2.plot()
Fail
pge2.check()
pge2.serialize()
.bin
MR750
Premier
MAGNUS
...
compile
EPIC interpreter source code
compile
GE simulator (WTools/Pulse Studio)

Interpreter (pge2)

Each Pulseq block is implemented directly as specified -- no interpolation.

Download: https://github.com/GEHC-External/pulseq-ge-interpreter

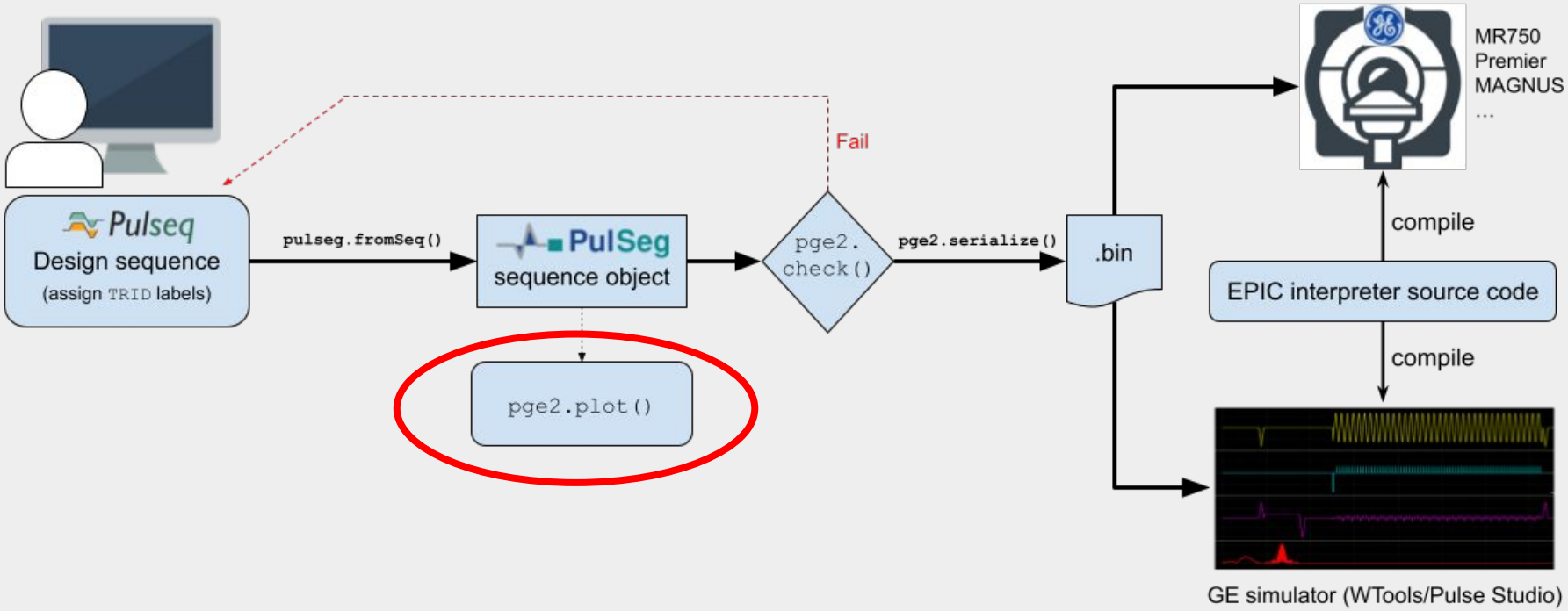
Freely shared within the GE research user community (MIT license)

For access, please contact your local GE representative

For questions, use the **Issues** and **Discussions** forums on Github

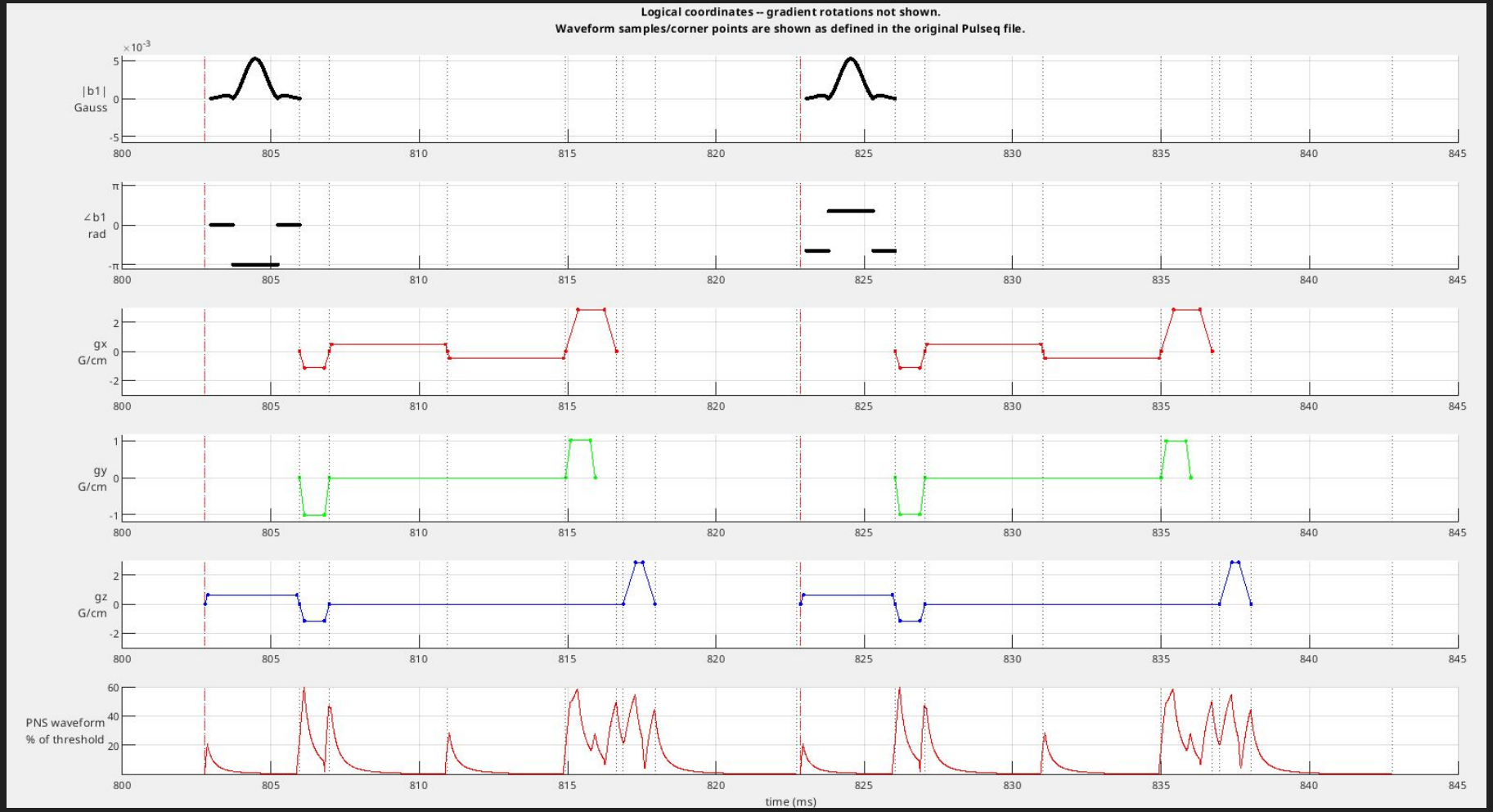
Refer to the **Wiki** for usage instructions, troubleshooting, and feature requests

Workflow

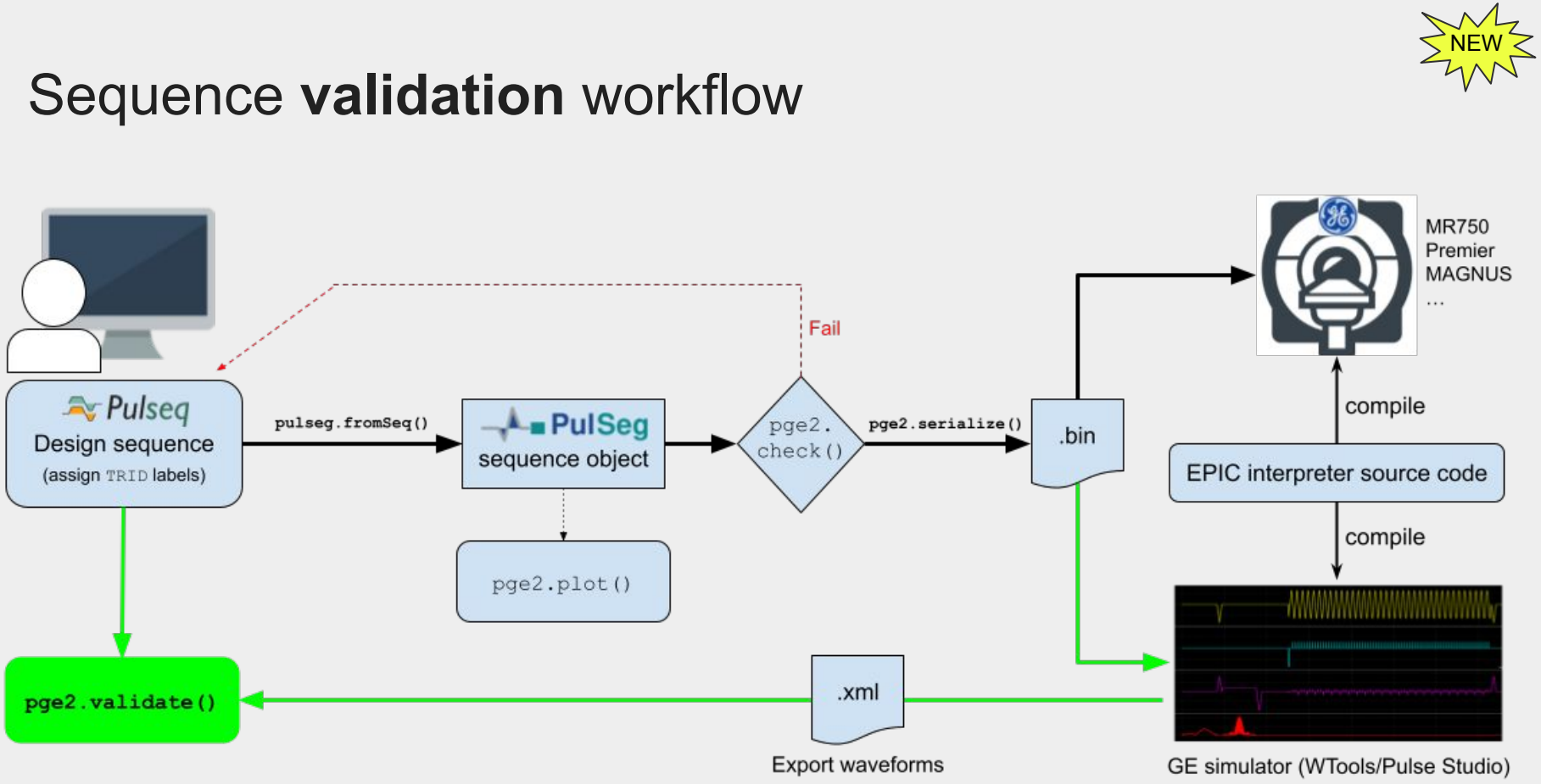

Pulseq
Design sequence
(assign TRID labels)
pulseg.fromSeq()
PulSeg
sequence object
pge2.plot()
Fail
pge2.check()
pge2.serialize()
.bin
MR750
Premier
MAGNUS
...
compile
EPIC interpreter source code
compile
GE simulator (WTools/Pulse Studio)

https://github.com/HarmonizedMRI/SequenceExamples-GE/tree/main/pge2

2D GRE



Sequence **validation** workflow



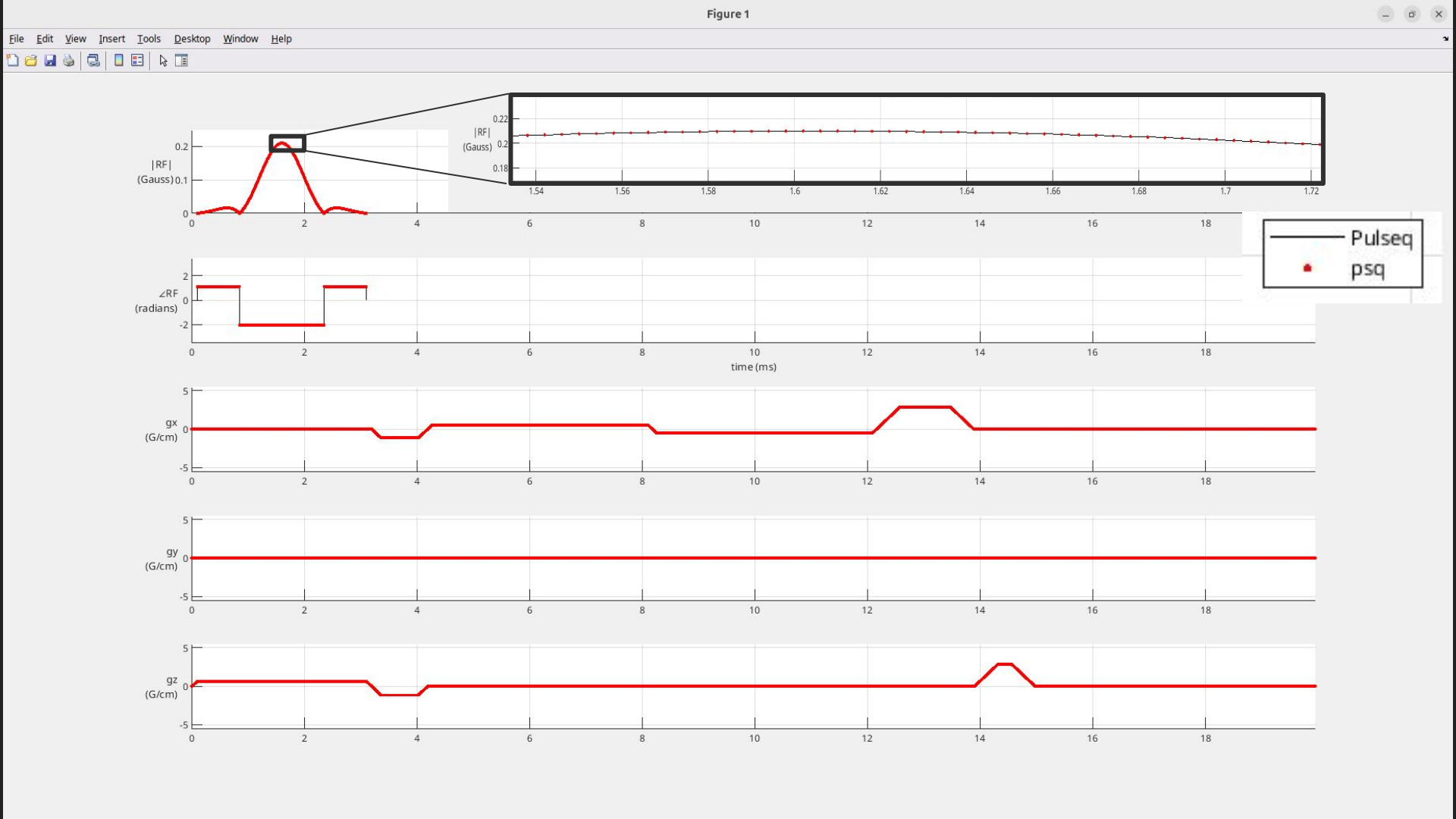


Figure 1
File Edit View Insert Tools Desktop Window Help
|RF| (Gauss)
∠RF (radians)
time (ms)
gx (G/cm)
gy (G/cm)
gz (G/cm)
Pulseq
psq

Low-Level Timing & Segment Rules

Timing rules:

- Rastes times: RF 2 us; Gradient /block duration 4 us; ADC 2 us or 1us (MAGNUS)
- Dead/ringdown times can often be ignored (block boundaries ‘disappear’)
- Gradient ramp duration >= 2 samples

Segment rules:

Fixed for all segment instances	Can vary between segment instances
Sequence of base blocks	RF/gradient amplitudes
Block duration (except pure delay blocks)	RF/gradient/ADC phase and frequency offsets
	Duration of pure delay blocks
	Gradient rotation (entire segment -- for now)

Resources

https://github.com/HarmonizedMRI/pulseq

https://github.com/HarmonizedMRI/pge2

Examples:
https://github.com/HarmonizedMRI/SequenceExamples-GE/tree/main/pge2

Interpreter source code: https://github.com/GEHC-External/pulseq-ge-interpreter

Further details on Thursday: “Patient safety and hardware Protection”